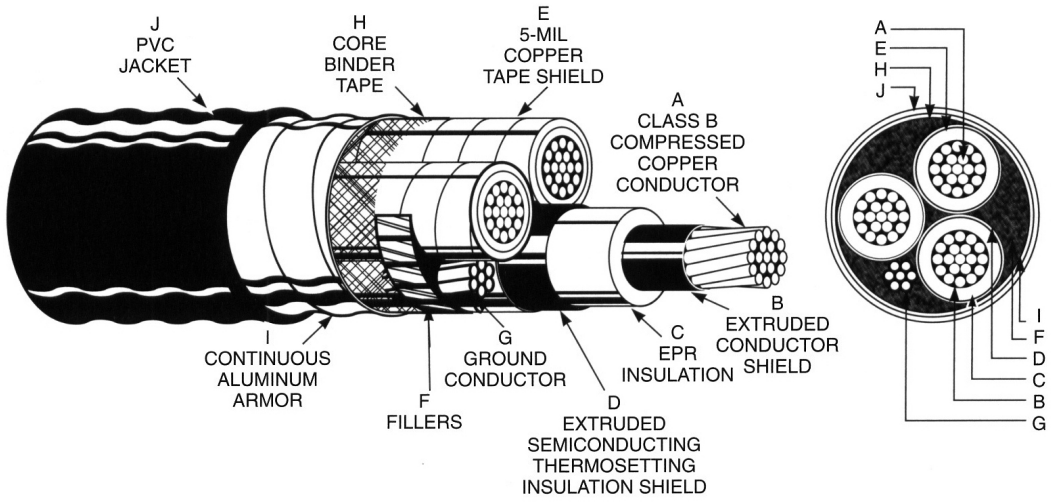
HLX13ET

ARMOR-X® -HL.
15kVU Shielded Type MV-105 and MC-HL.
EPR Insulated, 105°C.
Bare Copper Grounding Conductor.
Continuous Corrugated Aluminum Armor with -40°C PVC Jacket.



APPLICATIONS

Southwire HLX13ET ARMOR-X Type MV-105 or MC-HL provides an impervious sheath recommended as an economical alternative to traditional conduit systems. For use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs or continuous rigid cable supports. For use in Classes I, II, and III, Division 1 and 2 hazardous locations covered under NEC Articles 501, 502, and 503. These cables are capable of operating continuously in wet or dry locations at a maximum conductor temperature of 105°C for normal operation, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated 15,000 volts, 133% insulation level. Maximum sidewall pressure is 1000 lbs.

SPECIFICATIONS

Southwire HLX13ET Type MV-105 or MC-HL Cable is manufactured and tested in accordance with the latest revisions of the following standards and specification:

- UL 1072 - Medium Voltage Power Cables.
- UL 2225 - Safety Metal-Clad Cables and Cable-Sealing Fittings for Use in Hazardous (Classified) Locations.
- UL 1309 - Listed as Marine Shipboard Cable.
- ABS Listed as CWCMC.
- ICEA S-93-639 (NEMA WC 74) - 5-46 kV Shielded Power Cable for Use in the Transmission & Distribution of Electric Energy.
- IEEE 1202/FT4 - Flame Testing of Cable for Use in Cable Tray in Industrial and Commercial Occupancies with CPE and Solonon jacket(70,000 Btu/hr).
- ICEA T-29-520 - Vertical Cable Tray Flame Test (210,000 BTU/hr.).

Certified qualification tests were performed in accordance with the requirements of AEIC CS-8. Cable has fully met the qualification testing requirements of AEIC CS-8.

FEATURES

Southwire HLX13ET ARMOR-X Type MV-105 or MC-HL continuous corrugated aluminum armor provides a sheath impervious to moisture, liquids, and gases. Cable is flame retardant UL-listed for cable tray use, direct burial, sunlight resistant and minus 40°C installation temperature. Continuous aluminum sheath meets the grounding requirements of NEC 250-122 in non-HL locations. Cable provides proven, flexible EPR insulation. SOLONON™ low smoke, non halogen polyolefin jackets

Southwire
One Southwire Drive
Carrollton, Ga. 30119 USA

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and CPE jackets are available upon request.

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HLX13ET 3c, 15kV 220mil EPR(133%), TS, GW, Cont. Alum. Weld.

Product Code	Size	Conductor Diameter		0.220" (5.59mm) Insulation Diameter		Extruded Insulation Shield Diameter		Grd. Cond. Size	Approx. ARMOR-X Diameter		Jacket Thickness		Approximate Overall Diameter		Approximate Net Weight		Allowable Ampacities+	
	AWG or kcmil	inch	mm	inch	mm	inch	mm	AWG	inch	mm	inch	mm	inch	mm	lb./Mft.	kg/km	Direct Burial	In Air
HLX13ET-002	2	.283	7.19	.773	19.62	.858	21.78	6	2.29	58.17	.075	1.91	2.44	62	2540	3780	200	185
HLX13ET-010	1/0	.362	9.19	.853	21.65	.938	23.81	4	2.55	64.77	.075	1.91	2.7	68.6	3201	4763	255	240
HLX13ET-020	2/0	.405	10.29	.893	22.67	.978	24.83	4	2.55	64.77	.075	1.91	2.7	68.6	3533	5257	290	275
HLX13ET-040	4/0	.512	13	.998	25.34	1.053	26.73	3	2.75	69.85	.075	1.91	2.9	73.7	4663	6939	375	360
HLX13ET-250	250	.558	14.17	1.055	26.8	1.11	28.19	2	2.88	73.15	.075	1.91	3.03	77	5227	7778	410	400
HLX13ET-350	350	.661	16.79	1.158	29.4	1.243	31.56	2	3.22	81.79	.085	2.16	3.39	86.1	6809	10131	495	490
HLX13ET-500	500	.79	20.07	1.283	32.58	1.337	33.96	1	3.85	97.79	.085	2.16	4.02	102.1	8972	13351	590	600
HLX13ET-750	750	*.908	23.06	1.41	35.81	1.465	37.21	1/0	3.85	97.79	.085	2.16	4.02	102.1	11605	17268	720	745

+Ampacities are based on the NEC, 2011 Edition. Direct burial ampacities are based on table 310.83 three-conductors within an overall covering directly buried, 105°C conductor, 20°C earth ambient temperature. In air ampacities are based on table 310.71 three-conductors within an overall covering in free air, 105°C conductor, 40°C ambient temperature.
* Denotes compact stranding

SCOPE: This specification covers three-conductor EPR (ethylene propylene rubber) insulated, shielded, continuous corrugated aluminum armored, thermoplastic jacketed power cables with grounding conductor for use in aerial installations, direct burial, concrete-encased installations, cable trays, troughs, or continuous rigid cable supports. The cable will be listed Type MV-105 or MC-HL for use in Classes I, II, and III, Division 1 and 2, hazardous locations. These cables are capable of operating continuously at a temperature of 105°C for normal operations, 140°C for emergency overload conditions, and 250°C for short circuit conditions, and are rated 15,000 volts, 133% insulation level.

STANDARDS: The following standards will form part of this specification: UL Standard 1072 for Medium Voltage Power Cable and ICEA S-93-639 (NEMA WC74) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy.

CONDUCTOR: The conductor will be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8 and ICEA .

CONDUCTOR SHIELD: The conductor will be shielded with an extruded semiconducting thermosetting polymeric layer, which shall be firmly bonded to the insulation. The thickness shall be in accordance with the referenced standards.

INSULATION: The insulation will be EPR (ethylene propylene rubber) meeting the requirements of the referenced standards. The average thickness shall be 0.220".

INSULATION SHIELD: The insulation will be shielded with an extruded semiconducting thermosetting polymeric layer which will be identified as semiconducting. Over this layer will be applied a helically-wrapped 5-mil copper tape with 25% overlap. The method of phase identification will be similar to ICEA Method 3, using printed circuit numbers and colors (1-BLACK, 2-RED, 3-BLUE).

GROUND CONDUCTOR: The ground conductor will be Class B compressed concentric stranded bare copper in accordance with ASTM B3 and B8.

ASSEMBLY: The insulated and shielded power conductors will be cabled round with fillers and with a grounding conductor in one outer interstice and covered with a binder tape.

ARMOR: Continuous, impervious, welded, corrugated copper-free aluminum armor will be applied over the assembly.

JACKET: The cable will be covered with a red PVC jacket conforming to the requirements specified for polyvinyl chloride in ICEA. The average thickness will be in accordance with the referenced standards. The jacket will be sunlight resistant and will meet the requirements of the IEEE 1202 (70,000 Btu/hr) and ICEA T-29-520 (210,000 BTU/hr.) vertical cable tray flame tests. The jacket will be suitable for use at a minimum ambient temperature of minus 40°C. Optional SOLONON low smoke, non halogen polyolefin jackets and CPE jackets are available upon request.

IDENTIFICATION: Manufacturer's identification will be printed on the jacket.
TESTS: Qualificationtests shall be conducted in accordance with the requirements of AEIC.

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